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Electrochemical destruction of PFAS at low oxidation potential enabled by CeO_2 electrodes utilizing adsorption and activation strategies

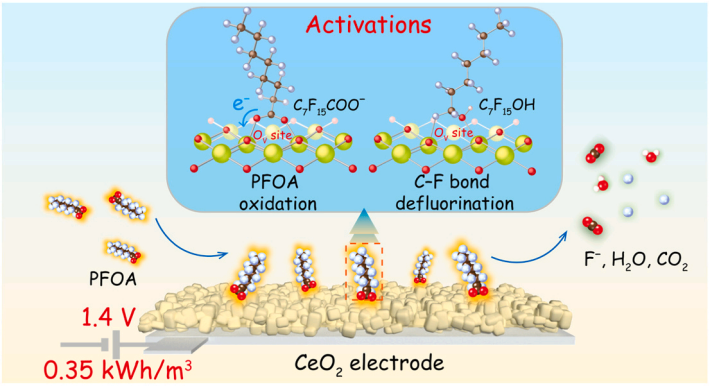
Xun Liu [a,b,1], Yufei Shu [b,1], Yu Pan [b], Guoshen Zeng [c], Meng Zhang [b], Chaoqun Zhu [b], Youmei Xu [b], Aling Wan [b], Mengxia Wang [a,b], Qi Han [b], Bei Liu [b], Zhongying Wang [b,d,*]

[a] School of Environment, Harbin Institute of Technology, Harbin 150086, PR China
[b] School of Environmental Science and Engineering, Southern University of Science and Technology, Shenzhen 518055, PR China
[c] School of Environmental Science and Engineering, Sun Yat-sen University, Guangzhou 510006, PR China
[d] Guangdong Provincial Key Laboratory of Soil and Groundwater Pollution Control, Southern University of Science and Technology, Shenzhen 518055, PR China

HIGHLIGHTS

- CeO_2 electrode achieves 94.0 % PFOA removal and 73.0 % defluorination efficiency.
- The process operates at a low oxidation potential, minimizing the conversion and interference of NOM and Cl^-.
- DFT analysis confirms that oxygen vacancies significantly reduce the energy barriers for defluorination.
- This method consumes only 0.62 kWh/m^3 per order, offering a highly energy-efficient solution for PFAS water remediation.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

The persistence and ecological impact of per- and poly-fluoroalkyl substances (PFAS) in water sources necessitate effective and energy-efficient treatment solutions. This study introduces a novel approach using cerium dioxide (CeO_2) electrodes enhanced with oxygen vacancy (O_v) to catalyze the defluorination of PFAS. By leveraging the unique affinity between cerium and fluorine-containing species, our approach enables adsorptive preconcentration and catalytic degradation at low oxidation potentials (1.37 V vs. SHE). Demonstrating high removal and defluorination efficiencies of perfluorooctanoic acid (PFOA) at 94.0 % and 73.0 %, respectively, our approach also proves effective in the environmental matrix. It minimizes the impacts of co-existing natural organic matter and chloride ions, crucial benefits of operating at lower oxidation potentials. The role of O_v in CeO_2 is validated by both experimental results and density functional theory modeling, demonstrating that these sites can activate the C–F bond and substantially reduce the energy barriers for defluorination. Consequently, our CeO_2-based method not only achieves defluorination efficiencies comparable to more energy-intensive techniques but does so

* Corresponding author at: School of Environmental Science and Engineering, Southern University of Science and Technology, Shenzhen 518055, PR China
E-mail address: wangzy6@sustech.edu.cn (Z. Wang).
[1] these authors contributed equally to this work.

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while requiring less than 0.62 kWh/m^3 per order. This positions our approach as a promising, cost-effective alternative for the remediation of PFAS-contaminated waters, emphasizing its relevance and effectiveness in environmental remediation scenarios.

1. Introduction

Poly- and per-fluoroalkyl substances (PFAS) play a crucial role in the global chemical industry, with widespread applications in hydrophobic coatings, anti-fouling treatments, moisture-resistant surfaces, and fire-extinguishing agents [1–3]. These substances are characterized by their remarkable stability and resistance to degradation, primarily due to the robust carbon–fluorine (C–F) bonds. This stability confers resistance to degradation in both wastewater treatment facilities and natural environments [4]. Their toxic effects on human health, including irreversible damage to liver function and the immune system, as well as their potential to cause cancer, pose significant risks [2]. Furthermore, PFAS can accumulate in biological tissues, resulting in bioaccumulation and potentially harmful impacts throughout the food chain [5]. Recently, increasing environmental awareness has cast a spotlight on the ecological impacts of PFAS release into natural ecosystems, leading to regulatory efforts aimed at curtailing their usage and fostering the development of environmentally benign alternatives [6,7]. Nevertheless, the enduring legacy of PFAS, which has accumulated over decades, remains evident in diverse environments, including the human bloodstream and polar regions [8–10]. The persistent presence of PFAS in these contexts highlights the critical and immediate need to address the issue of PFAS pollution.

To meet rising demands, environmental researchers globally have developed various methods for the degradation of PFAS, including photocatalysis [11,12], electrochemical oxidation processes [13,14], and alkaline hydrothermal treatment [15,16]. Among these, electrochemical oxidation has gained attention for PFAS degradation due to its ability to break down these contaminants into less harmful mineralized forms without the need for additional chemicals, thus minimizing secondary pollution [17,18]. Despite its potential, the method confronts considerable challenges impeding its broad implementation. One primary obstacle is the requirement for high anodic potentials at non-active anodes (e.g., Ti_4O_7, PbO_2, and boron-doped diamond (BDD)), often exceeding 3 V *vs* the Standard Hydrogen Electrode (SHE) [17,19]. Such conditions can inadvertently lead to the oxidation of co-existing non-toxic organics, such as natural organic matter (NOM), and provoke intensive water splitting [17]. Additionally, employing high oxidation potentials can convert chloride ions present in water into environmentally detrimental hypochlorite and chlorate ions [20]. These side reactions not only complicate the selective degradation of PFAS but also escalate energy consumption.

To overcome these inherent challenges associated with the electrochemical degradation of PFAS, employing parallel plate electrodes coated with materials that exhibit a high affinity for fluorine species represents a promising strategy. These materials are expected to attract and concentrate the PFAS on the electrode surface, and catalyze their degradation at lower potential through the activation of C–F bonds. This strategy has demonstrated considerable success in the reduction of CO_2 and NO_3^- by Cu-based materials [21–23], as well as in the oxidation of As (III) by Fe-based material [24,25]. However, currently, the electrode material specifically for the activation of C–F bond and catalytic degradation of PFAS is lacking.

The search for suitable electrode materials capable of activating C–F bonds can be informed by insights gained from C–F bond activation and cleavage in organic fluorine chemistry. Rare earth metals have shown significant potential and effectiveness due to their strong interactions with fluorine atoms, including the formation of metal–F bonds, which weaken C–F bonds and reduce their bond energy [26,27]. Among these metals, cerium (Ce)-based materials have gained prominence due to their abundance and the catalytic dynamics involving the interconversion between Ce(III) and Ce(IV) states [28,29]. Integrating Ce(III) into conventional electrodes, such as PbO_2 and Ti_4O_7 electrodes, has proven effective in enhancing electron transfer and thereby facilitating the degradation of PFAS [30,31]. However, the specific catalytic effects of Ce-based materials on PFAS degradation remain poorly understood and warrant further investigation to fully elucidate whether CeO_2 can achieve effective catalytic degradation of PFAS and to identify the catalytic active sites involved, thereby advancing its efficacy and potential application in the environmental remediation of PFAS.

In this study, we developed cerium oxide (CeO_2) electrodes integrated into a flow-by plate electrode system to investigate the adsorptive preconcentration and catalytic degradation of perfluorooctanoic acid (PFOA), a prevalent PFAS compound. We assessed the removal and defluorination efficacy of PFOA under various experimental conditions, including different voltages, pH levels, and electrolyte concentrations, and extended these evaluations to include simulated natural water matrices containing NOM and chloride ions. Additionally, we employed density functional theory (DFT) to elucidate the activation mechanisms involved in PFOA degradation at the electrode surface. Our experiments were broadened to test the electrode's effectiveness across a spectrum of PFAS types and concentration levels that more closely mimic real-world environmental conditions. A comparative analysis of the energy consumption of our system was also conducted, highlighting its potential application for efficient and energy-saving PFAS remediation.

2. Materials and methods

2.1. Chemicals

All chemicals used in this study were of reagent grade or higher and were used as received without any further purification. Cerium(III) chloride heptahydrate ($CeCl_3{\cdot}7\ H_2O$, 99 %), citric acid (99.5 %), sodium hydroxide (NaOH, 95 %), polyacrylonitrile (PAN, $[C_3H_3N]_n$, M.W. 250k), activated carbon (AC, 400 mesh), perfluoromethanesulfonic acid (PFOS, 99.5 %), perfluorononanoic acid (PFNA, 99 %), perfluorohexanoic acid (PFHxA, 99 %), perfluorobutanoic acid (PFBA, 99 %), and carbon black (CB) were purchased from Macklin, Shanghai. Sodium chloride (NaCl, 99 %) and sodium sulfate (Na_2SO_4, 99 %) were from Aladdin, Shanghai. The PFOA (97.6 %) was obtained from Dr. Ehrenstorfer GmbH (Germany). Fulvic acid (FA, average chemical formula of $C_{14}H_{12}O_8$, 85 %), employed as a representative of NOM, was purchased from Macklin, Shanghai. All aqueous solutions for the experiments were prepared using ultrapure water from Millipore, with a resistivity of 18 MΩ•cm.

2.2. Preparation of CeO_2-based electrodes

The synthesis of CeO_2 powder was conducted following a hydrothermal method described in previous literature [32], and the detailed procedure is supplied in Text S1, Supplementary Information. For the preparation of the functionalized anode electrodes, a homogeneous slurry mixture was composed of 120 mg of CeO_2, 7.5 mg of PAN as a binder, and 15 mg of carbon black to enhance conductivity. These components were uniformly dispersed in NMP solvent. This homogenized slurry was then carefully applied onto a 5 cm × 5 cm titanium plate, which acted as the substrate for the CeO_2 electrode. After coating, the electrode was dried in a vacuum oven overnight at 60 °C, ensuring the proper adhesion and stability of the CeO_2 coating. Concurrently, an AC electrode was prepared following a similar procedure, with the only

variation being the substitution of CeO_2 with 120 mg of AC in the slurry composition.

2.3. Catalytic degradation of PFAS

The degradation performance of CeO_2 electrodes against PFAS was evaluated using a recirculation system, incorporating a parallel plate electrode assembly, an external power supply (2231A-3003, KEITHLEY), and a peristaltic pump, as shown in Figure S1. The electrodes were positioned parallel to each other with a 3 mm gap. Within this configuration, the prepared CeO_2 electrode was utilized as the anode, while the AC electrode served as the cathode. For the degradation tests, we prepared a 50 mL solution containing 1 mg L^{-1} PFOA and 100 mg L^{-1} chloride ions (added through the dissolution of NaCl), simulating surface waters contaminated with perfluorinated compounds. The system operated with a steady flow rate of 10 mL/min, and an electrical voltage of 1.4 V was applied during the experiments unless specified otherwise. Sampling for analysis was conducted at various time intervals. Each 1 mL sample was bifurcated for separate analyses: 0.5 mL was dedicated to quantifying F^- and Cl^- concentrations through ion chromatography (IC; Dionex Aquion RFIC, Thermo Fisher Scientific), and the remaining 0.5 mL was analyzed for residual PFOA concentration using high-performance liquid chromatography-mass spectrometry (HPLC-MS; Q-Exactive, Thermo Fisher Scientific). The removal efficiency (R) of PFOA was determined using Eq. (1):

$$R_{PFOA} = (1 - \frac{C_{PFOA,t}}{C_{PFOA,0}}) \times 100\% \quad (1)$$

where $C_{PFOA,0}$ and $C_{PFOA,t}$ (mg L^{-1}) represent the initial and the remaining PFOA concentrations at the time t (min), respectively. Additionally, to evaluate the potential leaching of cerium during the degradation process, we employed inductively coupled plasma atomic emission spectroscopy (ICP-AES; iCAP 7400 series, Thermo Fisher Scientific) for monitoring. In experiments where 25 mg L^{-1} NOM was introduced to mimic environmental variables that could affect the degradation process, we assessed the changes in NOM concentration using ultraviolet-visible spectrophotometry (UV-Vis; Hitachi UH5300) at a wavelength of 254 nm [33].

To accurately quantify the transformation of various fluorine species during the PFOA removal process, a reverse bias of −1.4 V was applied for 3 h, facilitating the detachment of fluorine species from the CeO_2 electrode. During this phase, 50 mL of ultrapure water was employed as the receiving medium. To prevent potential reverse adsorption of fluorine species onto the AC electrode surfaces, a 5-min short circuit was initiated immediately following the reverse bias period. Subsequently, the concentrations of the desorbed PFOA and released F^- were quantitatively determined, and the potential degradation intermediates were identified through HPLC-MS, with the detailed methodology provided in Text S2 and S3. The nearly complete desorption of PFOA and F^- adsorbed on the CeO_2 electrode upon bias reversal has been confirmed, as shown in Figure S2. The removal efficiency of PFOA through electrosorption was then calculated using Eq. (2):

$$R_{electrosorption} = \frac{C_{PFOA,reverse}}{C_{PFOA,0}} \times 100\% \quad (2)$$

where $C_{PFOA,\ reverse}$ represents the concentration of desorbed PFOA in the regeneration solution (mg L^{-1}). Additionally, the defluorination efficiency (D) of PFOA was determined using Eq. (3):

$$D_{defluorination} = \frac{C_{F-reverse}}{C_{PFOA,0} \times \frac{15 \times M_F}{M_{PFOA}}} \times 100\% \quad (3)$$

where $C_{F\text{-reverse}}$ is the concentration of desorbed F^- in the regeneration solution (mg L^{-1}), M_F is the atomic mass of a fluorine atom (19 g/mol), and M_{PFOA} represents the molecular weight of a PFOA molecule (414 g/mol).

In the PFOS degradation test, the procedure mirrored that used for PFOA degradation, with the exception being the composition of the treatment solution, which contained 1 mg L^{-1} PFOS and 100 mg L^{-1} Cl^-. The desulfurization efficiency ($D_{sulfurization}$) was determined using Eq. (4):

$$D_{sulfurization} = \frac{C_{Sulfate-reverse}}{C_{PFOS,0} \times \frac{M_S}{M_{PFOS}}} \times 100\% \quad (4)$$

where $C_{\text{sulfate-reverse}}$ is the concentration of desorbed sulfate in the regeneration solution (mg L^{-1}), and M_S and M_{PFOS} represent the molecular weights of sulfate ion (96 g/mol) and PFOS (500 g/mol), respectively.

To evaluate the applicability in environmental matrix samples, surface water and groundwater were collected on the SUSTech campus using standardized sampling methods. On-site measurements of pH, conductivity, and dissolved oxygen were conducted using a multiparameter tester (AQUAREAD, AP-7000). The collected water samples were then immediately used to prepare 1 mg/L PFOA aqueous solutions for removal experiments. All remaining experimental procedures, including sample collection and analysis, followed the methods previously described.

2.4. Characterizations

The morphology of the CeO_2 material was examined using transmission electron microscopy (TEM; Talos F200X G2, Thermo Fisher), scanning electron microscopy (SEM; Merlin, Zeiss), and scanning transmission electron microscopy (STEM; Tecnai G2 F30 S-TWIN, Thermo Fisher). The X-ray powder diffraction (XRD; Rigaku Smartlab) was used to obtain the crystalline phase of the CeO_2 material. Atomic Force Microscopy (AFM) was employed to observe the surface smoothness of the electrode. Water contact angle measurements were conducted using the DAS25E instrument from KRUSS to analyze the hydrophilicity of the electrode surface. X-ray photoelectron spectroscopy (XPS; PHI 5000 VersaprobeIII, ULVAC-PHI) was employed to analyze the chemical composition of CeO_2 and compositional changes of PFAS on the electrode surface during electrochemical treatments. Additionally, an Electron Paramagnetic Resonance Spectrometer (EPR, EMXPlus-10/12) was employed to monitor the generation of free radicals during the electrochemical treatment process.

3. Results and discussion

3.1. Characterization of CeO_2 material and electrode

The CeO_2 was synthesized through the thermal annealing of $Ce(OH)_3$. This transformation was evident when comparing the XRD patterns of $Ce(OH)_3$ and CeO_2 (Figure S3a). XPS spectra revealed a predominance of cerium in its tetravalent state, with about 15 % existing in the trivalent state (Figure S3b), suggesting a significant presence of oxygen vacancies, consistent with previous research findings [34]. The detailed morphology and structure of the CeO_2 material is shown in Figure S4, which reveals that the synthesized CeO_2 exhibited an irregular morphology with an average particle size of approximately 50 nm. The electrode coated with CeO_2 exhibited a smooth and hydrophobic surface, characterized by the AFM surface scanning and a water contact angle of approximately 104° (Figure S5). Such hydrophobic characteristics of the electrode surface are anticipated to facilitate the adsorption of PFAS molecules [35].

3.2. Catalytic degradation of PFOA

To enhance mass transfer and PFOA degradation, a flow-by plate electrode system was adopted to concentrate PFOA on the electrode

surface by adsorption. The degradation efficiency of PFOA by CeO_2 electrodes over time and across different applied voltages is illustrated in Fig. 1a. Remarkably, even without an applied voltage (0 V), the electrode was capable of removing about 34.7 % of PFOA within 180 min. The introduction of external voltages significantly boosted the removal efficiency, with observed efficiencies reaching 49.7 %, 68.3 %, 78.2 %, and 73.7 % at voltages of 0.8 V, 1.2 V, 1.4 V, and 1.6 V, respectively. An interesting observation was the absence of fluoride ions in the solutions under these conditions, as shown in Figure S6. However, this changed upon applying a reverse bias of –1.4 V, where both PFOA and fluoride ions were detected in the receiving solutions (i.e., ultrapure water), suggesting the electrosorption of PFOA and fluoride ions on the surface of the CeO_2 electrode. This phenomenon can be attributed to the strong affinity of cerium-based materials for fluorine species, including both PFAS and fluoride ions [36,37]. As a control experiment, the BDD electrode and PbO_2 electrode were tested under the same experimental conditions but did not achieve any degradation of PFOA, as shown in Figure S7.

To elucidate the individual contributions of degradation and adsorption to PFOA removal, the proportions at various voltages were determined via Eqs. (1) and (2) and are illustrated in Fig. 1d. Notably, at 0 V, where no voltage was applied, the defluorination efficiency was negligible, indicating that PFOA was primarily adsorbed onto the electrode without undergoing any defluorination. The highest defluorination efficiency, 57.2 %, occurred at an applied voltage of 1.4 V. At other voltages, efficiencies were recorded at 5.4 % at 0.8 V, 37.6 % at 1.2 V, and 51.5 % at 1.6 V. The reduced efficiency at lower voltages can be due to slower adsorption and oxidation rates of PFOA on the electrode. Conversely, the lower efficiency at 1.6 V may be linked to the onset of cerium oxidation, transitioning from Ce(III) to Ce(IV), as shown in Figure S8. This oxidation potentially alters the active sites on the CeO_2 electrodes, affecting their catalytic activity for PFOA degradation. Consequently, an applied voltage of 1.4 V was identified as the optimal condition to maximize defluorination efficiency in subsequent experiments unless otherwise specified.

To determine the oxidation potential of the CeO_2 electrode in our dual-electrode system, we employed an Ag/AgCl electrode as the reference. The detailed procedure and calculations can be found in Text S4 [24]. The working potential of the CeO_2 electrode relative to the Ag/AgCl electrode was calculated with an applied bias ranging from 1.0 to 1.6 V, resulting in oxidation potentials in the range of 0.89 V to 1.36 V vs. Ag/AgCl (equivalent to 1.09–1.56 V/SHE), as shown in Figure S9. Notably, these oxidation potentials are significantly lower than those typically required for conventional, non-active electrodes [19,38]. To confirm the effectiveness of this low potential in oxidizing PFOA, linear sweep voltammetry scans were conducted in a three-electrode system. As shown in Figure S10, the CeO_2 electrode exhibited a distinct oxidation peak starting above 0.9 V vs. Ag/AgCl (equivalent to 1.1 V/SHE) in a solution containing PFOA, and such a peak was absent in the PFOA-free electrolyte, suggesting that this peak is associated with PFOA oxidation. The low oxidation potential observed in both the three-electrode and the dual-electrode systems implies the catalytic effects of the CeO_2 electrode in lowering the oxidation potentials required for PFAS degradation.

To investigate the impact of environmental variables such as electrolyte concentration and pH on the removal and defluorination of PFOA, a series of experiments were carried out with varying chloride ion concentrations and pH levels. The absence of chloride ions yielded a modest PFOA removal efficiency of 61.2 % (Fig. 1b). The introduction of chloride ions at low concentrations, simulating the low conductivity found in natural groundwater and surface waters (typically less than 2 mS/cm) [17], significantly enhanced both the kinetics and efficiency of PFOA removal and defluorination. Specifically, increasing chloride ion concentrations to 100 mg L^{-1}, resulting in a conductivity of 0.035 mS/cm, and 250 mg L^{-1}, corresponding to 0.083 mS/cm, escalated the removal efficiency to 78.2 % and 83.2 %, respectively. Concurrently, defluorination efficiencies improved to 57.2 % and 65.6 % (Fig. 1e). The rise in solution conductivity increases the anode potential and facilitates the mass transfer, likely enhancing the removal efficiency. These results demonstrate the system's effectiveness in conditions of low electrolyte concentration, distinguishing it from traditional systems that depend on high electrolyte levels and specific electrolyte types (e.g., sulfate ion).

The impact of pH on the removal and defluorination of PFOA was systematically evaluated across a pH range from 4.0 to 8.1, reflecting the typical pH levels of surface and groundwater environments. Our results indicate that removal efficiency was notably lower at the higher pH of 8.1, achieving only 65.7 % (Fig. 1c). In comparison, at lower pH levels, removal efficiencies were significantly enhanced, consistently approximating 80 %, with defluorination efficiency varying slightly from 50.0 % to 57.1 %. This was a marked improvement over the 24.0 % defluorination observed at pH 8.1. This dependency of efficiency on pH is consistent with observations from electrooxidation studies of PFAS,

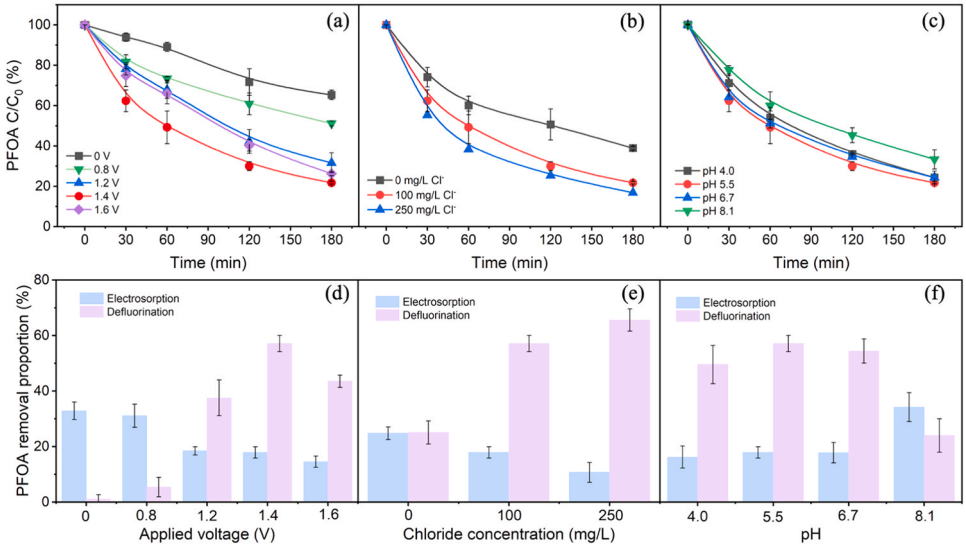


Fig. 1. Impact of various factors on the removal kinetics of PFOA and their proportions through defluorination and electrosorption: (a, d) applied voltage, (b, e) electrolyte concentration, and (c, f) solution pH. Standard reaction conditions unless noted otherwise: [Bias]= +1.4 V, pH = 5.5, [PFOA] = 1 mg L^{-1}, and [Cl^-] = 100 mg L^{-1}. The error bars represent the standard deviation of three replicate experiments.

which have documented a decrease in the generation of hydroxyl radicals at elevated pH levels; these radicals are crucial for the effective degradation of PFAS [39,40]. Considering the prevalent occurrence of PFAS in various water sources like drinking water, groundwater, and secondary treated wastewater effluents, which are often characterized by moderate pH and low electrical conductivity [17,41], our findings are particularly relevant.

Furthermore, our study expanded to monitor the transformation of fluorine species over time and calculate the fluorine mass balance, as detailed in Text S5. Concurrently, we tracked the removal kinetics and potential oxidation of co-existing NOM and chloride ions, given their documented interference in the electrooxidation removal of PFAS [42–45]. Fig. 2a illustrates the preferential removal of PFOA compared to NOM and chloride ions with a PFOA removal efficiency of ~94 % achieved within 12 h. In contrast, the CeO_2 electrode exhibited minimal interaction with chloride ions, displaying a removal efficiency of less than 10 % and rapidly reaching equilibrium. In the case of NOM, around 50 % was removed, primarily through adsorption rather than oxidative degradation, as evidenced by an 84.0 % desorption efficiency during regeneration (Figure S11). The capacity of our system to minimize interference from NOM and chloride ions holds significant implications for the electrooxidation treatment of PFAS. Traditional anodes require anodic potentials above 3 V/SHE to break the C–F bonds in PFAS [17,19]. At such high potentials, anodic oxidation of chloride ions and NOM can lead to increased energy consumption and potential environmental and health risks [46]. Opting for a lower voltage of 1.4 V, corresponding to an anodic potential of 1.37 V/SHE (Figure S8), our approach notably avoids the oxidation of chloride ions and NOM, mitigating the risk of generating unwanted by-products. In addition, degradation experiments conducted under varying NOM concentrations demonstrated that the system can effectively and selectively degrade PFOA in the presence of moderate levels of NOM, as shown in Figure S12.

Fig. 2b depicts the fluorine mass balance throughout the electrosorption-degradation process. As the treatment progressed, the proportion of residual PFOA in the solution consistently diminished, in line with the ongoing removal process. Meanwhile, the fraction of PFOA adsorbed on the electrode surface remained relatively stable at ~ 15 %. Notably, there was a significant increase in the fraction of fluoride ions identified as the final degradation products, escalating from 20 % at 1.5 h to 73 % by the 12-hour mark. Remarkably, this high level of

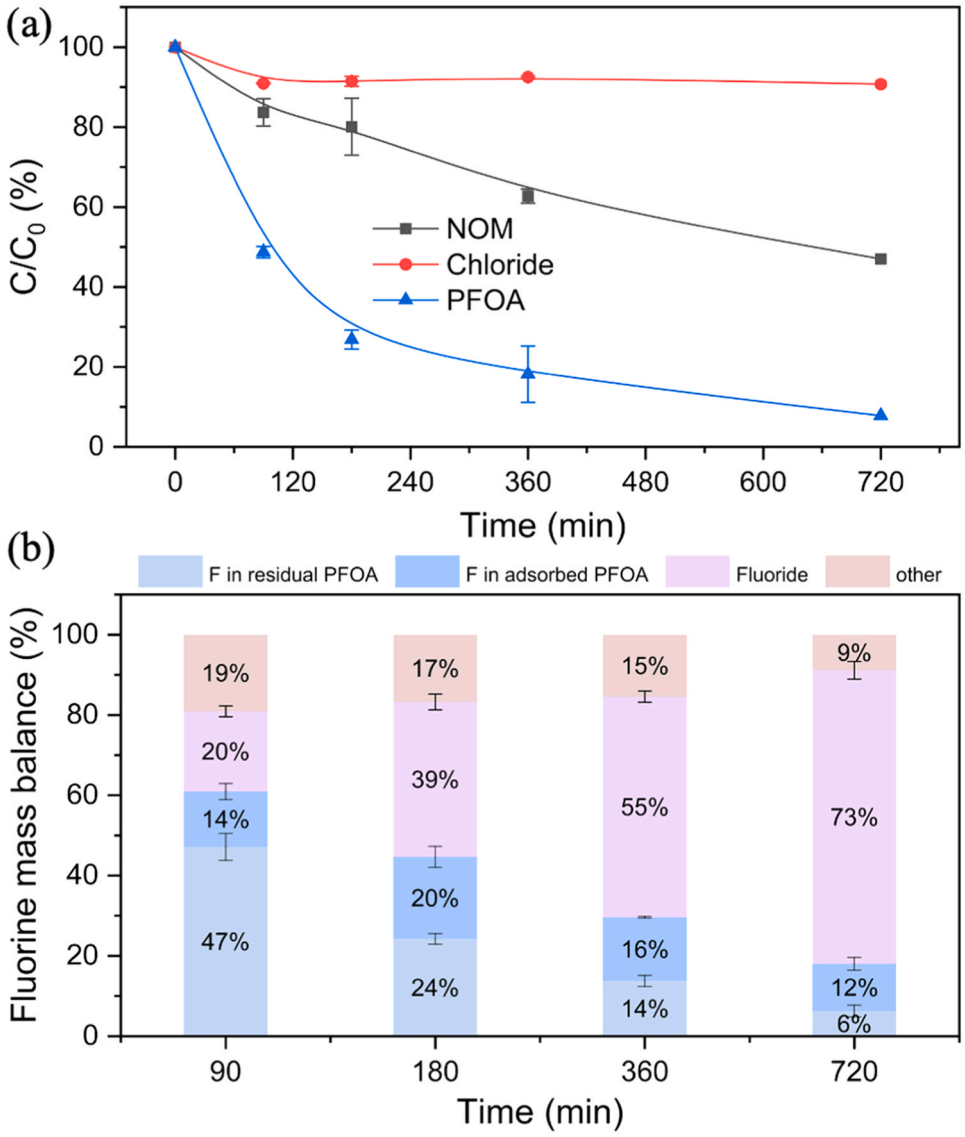


Fig. 2. Selective removal performance of CeO_2 electrode: (a) removal efficiency of PFOA, NOM, and Cl^-; (b) the fluorine mass balance over different time intervals during the treatment process. The category "Other" includes fluorine present in intermediate products and also accounts for experimental error. [Bias]= +1.4 V, pH = 5.5, [PFOA] = 1 mg L^{-1}, [Cl^-] = 100 mg L^{-1}, and [NOM]= 25 mg L^{-1}.

defluorination efficiency was achieved despite the use of a relatively low potential in our system, demonstrating effectiveness comparable to that achieved with higher-potential electrooxidation systems [47,48].

In the fluorine mass balance analysis, a notable pattern emerged involving a fraction of fluorine that could not be definitively categorized as either unreacted PFOA or as fluoride ions. This ambiguous fraction, labeled as "Other" in Fig. 2b, likely includes fluorine species adhering to device surfaces, as well as any intermediates formed during the degradation process. Notably, the proportion attributed to this "Other" category declined from 20 % at 1.5 h to 9 % after 12 h of treatment. This reduction suggests that fluorine-containing intermediates comprise a considerable portion of the "Other" fraction. These intermediates can pose additional water pollution risks if not further treated or mineralized, as their toxicity and persistence can vary [49]. Our method, however, effectively adsorbs fluorine-containing species during processing, avoiding the release of these intermediates and thereby helping to mitigate the potential environmental risks they may pose. This pattern also confirms the dynamic defluorination process within our system, signifying that PFAS compounds were progressively decomposed into intermediates, which eventually converted to fluoride ions, facilitated by the low-voltage, CeO_2-enabled electrochemical treatment approach.

3.3. The mechanism of catalytic PFOA degradation

Fig. 3 presents the TEM image and elemental mapping of the CeO_2 electrode used after a 3-h treatment at the applied voltage of 1.4 V. The images utilize distinct color codes to distinguish various components of the electrode: carbon black, which acts as a conductive agent, is encircled in red, while CeO_2 is marked in green. The elemental mapping analysis demonstrates that the distribution of fluorine species aligns more closely with cerium than with carbon. This observation strongly supports the pivotal role of CeO_2 in the adsorption and degradation processes of PFOA, underscoring its substantial influence on the degradation mechanism of PFOA.

To confirm the chemical transformation of fluorine species on CeO_2 electrodes under electrochemical treatment, we analyzed the time-resolved F 1 s XPS spectra of samples treated at 1.4 V compared to those without electrooxidation treatment (Figs. 3b and 3c), as detailed in Text S6. In the absence of an applied bias, a noticeable increase in the intensity of the peak at 687.3 eV was observed, which is characteristic of organic fluorine in the C–F bond. This increase supports the continuous adsorption and accumulation of PFOA on the CeO_2 electrode surface. Conversely, under electrochemical treatment at 1.4 V, the peak indicative of organic fluorine was initially present but diminished over time. Meanwhile, the peak at ~682.0 eV, corresponding to the fluoride ion, became more pronounced as the treatment progressed [50]. This shift in peak intensity reflects an almost complete conversion of adsorbed PFOA into fluoride ions on the CeO_2 electrode during electrochemical treatment.

To elucidate the mechanisms underlying the transformation of PFOA into F^- at low oxidation potentials, we employed DFT to analyze the adsorption and defluorination processes of PFOA interacting with CeO_2 (Text S7). Given the critical role of oxygen vacancies (O_V) in CeO_2 for catalytic reactions [51,52] and other materials for PFAS degradation [53,54], we studied the interaction behaviors of PFOA molecules on the CeO_2 (111) surface and at oxygen-defect sites of CeO_2 (Figure S13 and Table S2). The adsorption of PFOA via the carboxyl group at the O_V site (C_7F_{15}-COO^--CeO_2-O_V) exhibited significantly lower adsorption energy of –66.0 kJ/mol compared to 23.5 kJ/mol on the (111) surface without O_V (C_7F_{15}-COO^--CeO_2). This lower energy indicates a spontaneous preference for PFOA adsorption at the O_V site over the regular oxygen-exposed surfaces. Additionally, the adsorption of PFOA via the fluorine chain at the O_V site (COO^--C_7F_{15}-CeO_2-O_V) was less favorable, showing a positive adsorption energy of 101.9 kJ/mol, in contrast to the more favorable adsorption via the carboxyl group. The less favorable adsorption via the fluorine chain can be attributed to the steric hindrance, where the relatively long fluorine chain exceeds the spatial confines of the oxygen vacancy (Figure S13 d–e). Furthermore, the adsorption of the defluorinated intermediates (e.g., $C_6F_{13}COOH$) via the carboxyl group at the O_V site (C_6F_{13}-COOH-CeO_2-O_V) also exhibited a negative adsorption energy of −62.7 kJ/mol, suggesting that favorable adsorption could persist throughout the defluorination process.

The direct adsorption of the carboxyl group on the CeO_2 electrode facilitates the electron loss of PFOA through direct electron transfer to the electrode (Eq. 5). This process is supported by a significantly lower energy difference (ΔE) required to lose an electron when $C_7F_{15}COO^-$ is adsorbed by CeO_2-O_V (88.1 kJ/mol) compared to without CeO_2 (488.7 kJ/mol, Fig. 4a). The high energy associated with losing an electron without CeO_2, in line with values reported in previous DFT calculations [55], is attributed to the electron-withdrawing effect of multiple fluorine atoms in the PFOA molecule. This effect typically necessitates a higher oxidation potential in conventional electrooxidation treatments. However, this electron deficiency is partially offset when $C_7F_{15}COO^-$ is adsorbed on CeO_2-O_V, as evidenced by the charge density difference (CDD) plots (Fig. 4b). This interaction significantly reduces the energy required for electron loss, thereby potentially lowering the oxidation potential required for effective PFOA degradation. Following

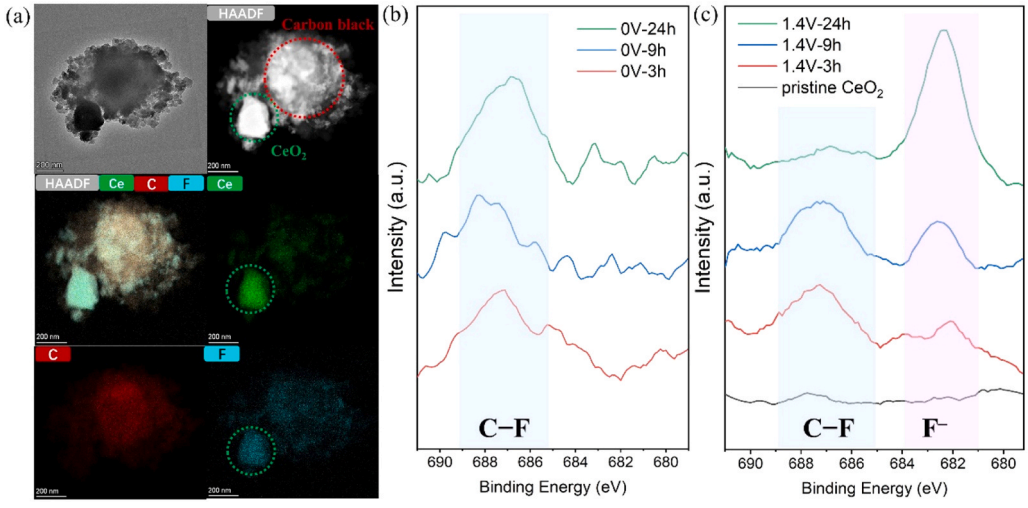


Fig. 3. Characterization of CeO_2 electrode post-PFOA degradation: (a) TEM image of CeO_2 electrode, along with elemental mappings for Ce, C, and F; time-resolved F 1 s XPS spectra of the used CeO_2 electrode under 0 V (b) and at the applied voltage of 1.4 V (c).

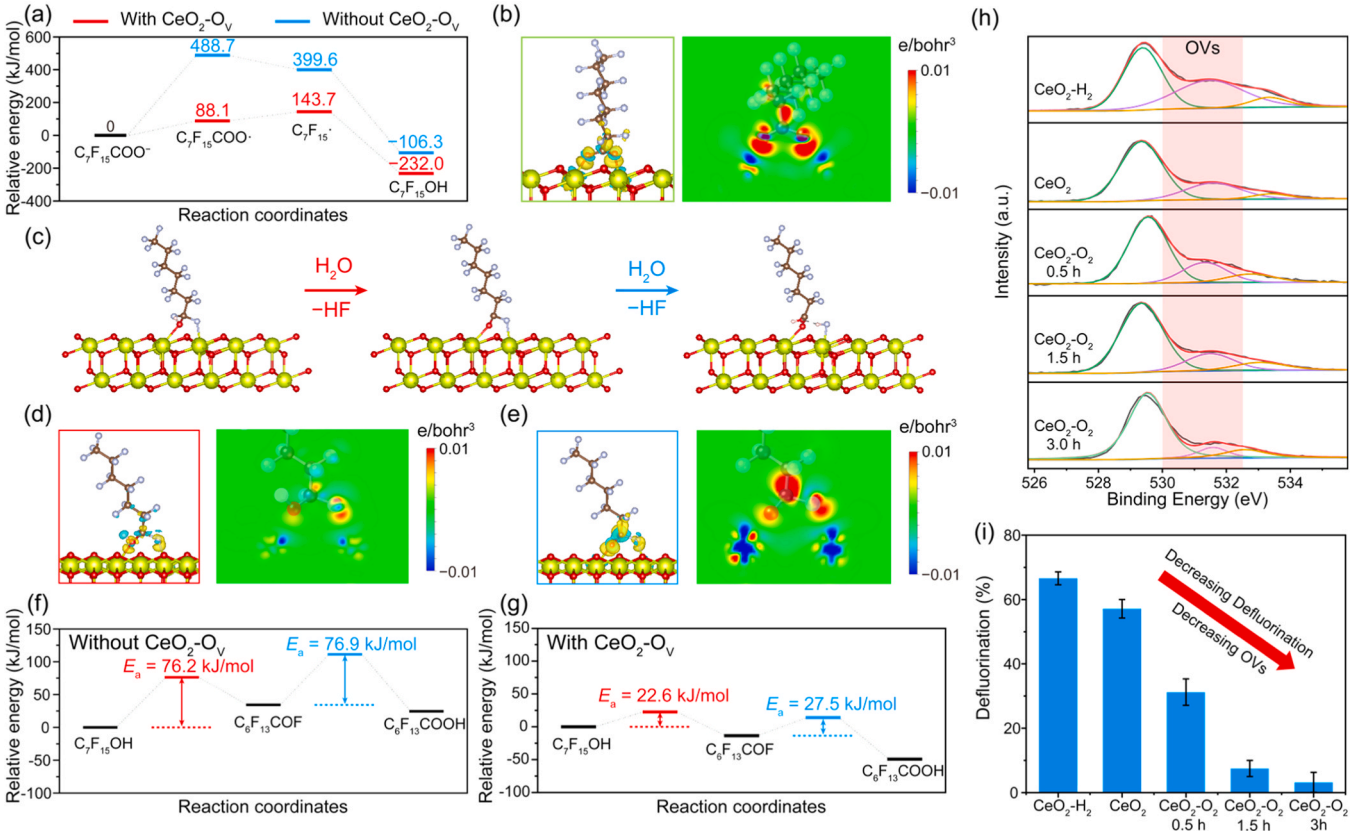


Fig. 4. DFT calculations of C–F activation and defluorination processes: (a) reaction coordinates of the PFOA oxidation process with and without CeO_2-O_V; (b) isosurface 3D plot of the CDD of PFOA when adsorbed on the CeO_2-O_V surface, with the right side panel showing CDD contour plot in the (010) cut plane; (c) defluorination steps for $C_7F_{15}OH$ and $C_6F_{13}COF$ adsorbed on the CeO_2-O_V surface; (d–e) isosurface 3D plots of the CDD of $C_7F_{15}OH$ (d) and $C_6F_{13}COF$ (e) when adsorbed on the CeO_2-O_V surface, with the right side of each panel displaying CDD contour plots in the (100) cut plane; (f–g) reaction coordinates for the defluorination of free intermediates (d) and the intermediates adsorbed on the CeO_2-O_V surface (e); (h) O 1 s XPS spectra and (i) defluorination efficiency of CeO_2 samples. In the CDD plots, regions of electron accumulation and depletion are indicated in red-yellow and cyan-blue, respectively, with an isosurface level set at 0.003 e/Bohr^3.

electron loss, the subsequent step is proposed to be the decarboxylation (Eq. 6), resulting in the formation of carbon radicals ($\cdot C_nF_{2n+1}$) [44], which were confirmed in this study by HPLC-MS results (Figure S14). These radicals can then react with hydroxyl radicals (•OH) to yield perfluorooctanol ($C_7F_{15}OH$, Eq. 7), with the generation of •OH validated by EPR measurements (Figure S15). Both steps proceed favorably according to our DFT calculations (Fig. 4a) and align with previous studies [55,56].

$$C_7F_{15}COO^- - e^- \rightarrow C_7F_{15}COO\bullet \quad (5)$$

$$C_7F_{15}COO\bullet \rightarrow C_7F_{15}\bullet + CO_2 \quad (6)$$

$$C_7F_{15}\bullet + \bullet OH \rightarrow C_7F_{15}OH \quad (7)$$

$$C_7F_{15}OH \rightarrow C_6F_{13}COF + HF \quad (8)$$

$$C_7F_{15}OH + H_2O \rightarrow C_6F_{13}COF + HF + H_2O \quad (9)$$

$$C_6F_{13}COF + H_2O \rightarrow C_6F_{13}COOH + HF \quad (10)$$

The defluorination steps following the generation of $C_7F_{15}OH$ and $C_6F_{13}COF$ are known to encounter high energy barriers due to the difficulty in breaking the robust C–F bonds (Eqs. 8–10) [57,58]. However, our DFT analysis reveals significant interactions between fluorine atoms with the surrounding Ce sites near the oxygen vacancy when these fluorine-containing intermediates are adsorbed on the CeO_2-O_V surface (Fig. 4c). Specifically, for $C_7F_{15}OH$, CDD plots demonstrate that one fluorine atom bonded to the α-C (C–F-1) interacts with a nearby Ce atom at the oxygen vacancy, while the other fluorine (C–F-2) does not contact any Ce atom due to steric hindrance (Fig. 4d). This interaction results in a significant increase in charge density around the C–F-1 bond and a decrease at the Ce interaction site. Consequently, the bond dissociation energy of the C–F-1 bond is reduced to 137.5 kcal/mol compared to 148.6 kcal/mol for the free $C_7F_{15}OH$ molecule, along with a slight increase in bond length from 1.38 Å to 1.40 Å, as detailed in Figure S16 and Table S3. In contrast, the changes in the bond dissociation energy and bond length of C–F-2 are minimal, implying that the activation of the C–F bond primarily occurs where the fluorine atom interacts with Ce.

The activation of C–F-2 becomes more pronounced in the subsequent defluorination of the intermediate $C_6F_{13}COF$ due to reduced steric effects, as supported by the changes in electron density around C–F-2 (Fig. 4e). This activation results in an elongated bond length (1.45 Å vs. 1.37 Å) and a decreased bond dissociation energy (119.6 kcal/mol vs. 144.3 kcal/mol) of the C–F-2 bond relative to the free $C_6F_{13}COF$ molecule, detailed in Figure S17 and Table S4.

The climbing image nudged elastic band (CI-NEB) method was employed to calculate the energy barriers for the defluorination processes of PFOA derivatives. The initial loss of a fluorine atom from $C_7F_{15}OH$ via elimination (Eq. 8) is thermodynamically challenging, as evidenced by a very high activation barrier (E_a) of 328.3 kJ/mol determined from transition state (TS) searching. In contrast, the hydrolytic defluorination pathway of $C_7F_{15}OH$ (Eq. 9) presents a lower E_a of 76.2 kJ/mol (Fig. 4f). This calculated energy barrier is consistent with typical values reported in the literature, which range from 55.92 to

88.9 kJ/mol [56,57], underscoring that defluorination is kinetically inhibited.

However, when $C_7F_{15}OH$ is adsorbed on the CeO_2-O_V, the E_a for the hydrolytic defluorination of the activated C–F-1 bond decreases significantly to 22.6 kJ/mol, as shown in Fig. 4g. Similarly, for the defluorination of $C_6F_{13}COF$ (Eq. 10), the E_a reduces from 76.9 kJ/mol for the free molecule to only 27.5 kJ/mol when facilitated by the interaction with CeO_2-O_V (Figs. 4f, 4g). Additionally, the defluorination processes of $C_7F_{15}OH$ and $C_6F_{13}COF$ are more energetically favorable when adsorbed on CeO_2-O_V, with changes in energy of −13.5 kJ/mol vs. + 34.3 kJ/mol and −35.7 kJ/mol vs. −9.8 kJ/mol, respectively, relative to the free derivative molecules (Figs. 4f, 4g). The overall energy scheme, which includes oxidation, decarboxylation, and defluorination of PFOA, is depicted in Figure S18. These results highlight the critical role of CeO_2 with oxygen vacancies in activating the C–F bond and lowering the energy barriers for the efficient defluorination of PFOA.

To validate the role of oxygen defects predicted by DFT and figure out the relationship between oxygen defects and PFOA destruction, we compared the performance of CeO_2 electrodes with varying concentrations of oxygen defects in removing and degrading PFOA. We prepared samples with high and low concentrations of defects by annealing CeO_2 in the H_2 and O_2 atmospheres, respectively (Text S1), according to established methods [59]. High-resolution O 1 s spectra in Fig. 4h, validated the successful synthesis of CeO_2 with varying defect concentrations. The samples were denoted as CeO_2-H_2 (defect-rich) and CeO_2-O_2-with different treatment times (oxygen-rich, essentially defect-less). The PFOA removal efficiencies of both CeO_2-H_2 and standard CeO_2 were similar and notably superior to that of CeO_2-O_2. The

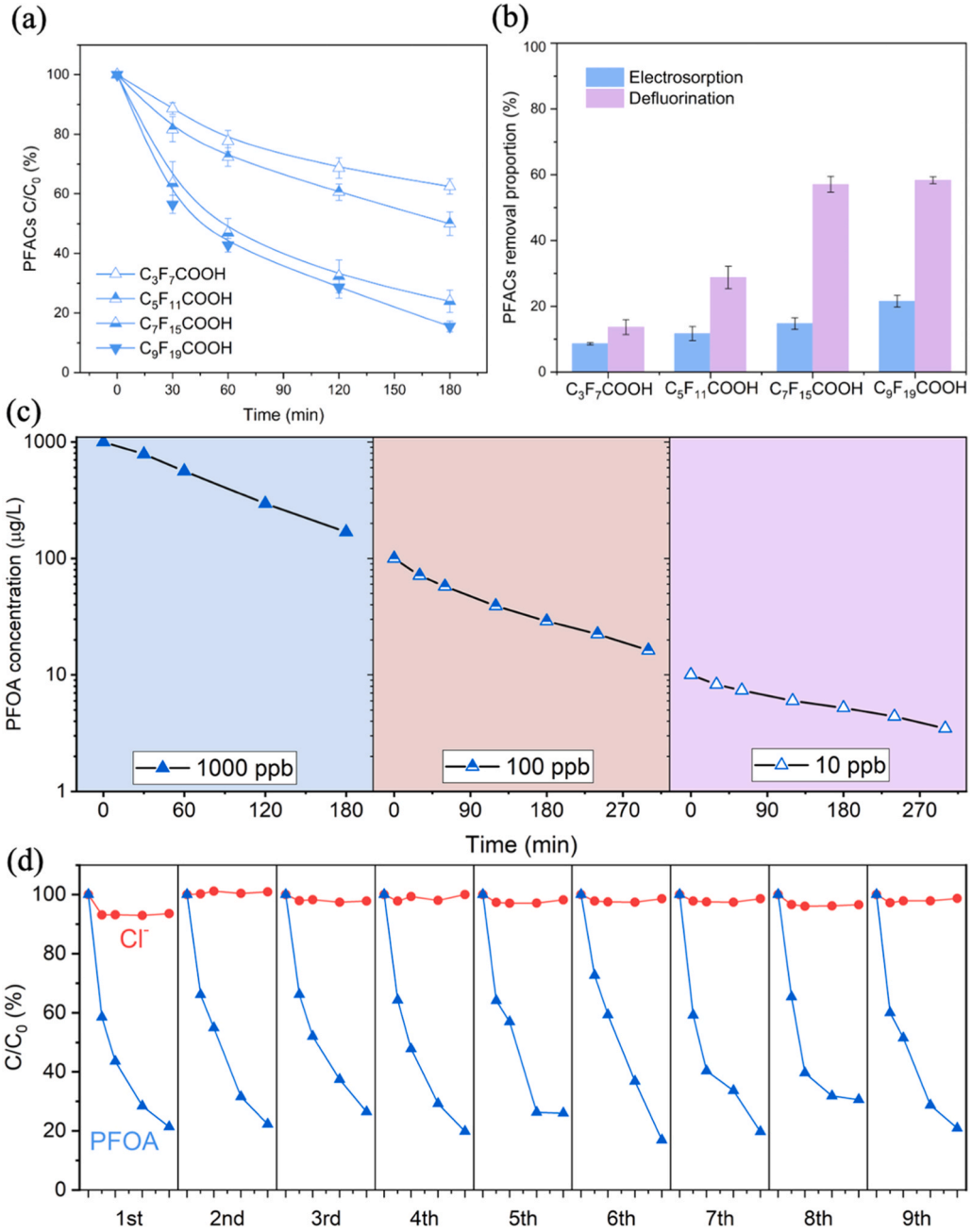


Fig. 5. (a) Performance across various carbon-chain PFACs; (b) corresponding electrosorption and defluorination percentages in the PFAC removal; (c) removal kinetics across different PFOA concentrations; and (d) selective PFOA removal over chloride ions in nine sequential runs. Each complete cycle consists of a 180-minute treatment process at 1.4 V followed by a 180-minute regeneration process at –1.4 V.

discrepancies in defluorination efficiency were even more pronounced in Fig. 4i and S19: CeO_2 with a moderate level of defects demonstrated a high defluorination efficiency of 57.3 %. Further increasing the defect concentration in CeO_2-H_2 led to the highest defluorination efficiency, achieving 66.6 %. In contrast, the defluorination efficiency of CeO_2-O_2 decreased significantly with prolonged calcination time in an oxygen atmosphere. Specifically, the defluorination rates were 31.2 % for CeO_2-O_2-0.5 h, 7.5 % for CeO_2-O_2-1.5 h, and 3.2 % for CeO_2-O_2-3h. These results effectively validate our DFT findings that oxygen defects in CeO_2 enhance the exposure of cerium sites. This exposure facilitates interaction with adsorbed PFOA and its degradation intermediates, leading to highly efficient catalytic degradation.

3.4. Practical applicability of CeO_2-enabled electrooxidation treatment

To conduct a more comprehensive evaluation of the electrode's versatility, robustness, and practical applicability, we conducted a series of experiments to assess its performance across a range of PFAS, varying concentration levels, and its long-term cycling stability. Fig. 5a showcases the efficacy of the CeO_2 electrode against four perfluorinated carboxylic acids with varying chain lengths. The results indicated that the electrode was more effective at removing longer-chain PFAS: defluorination efficiencies were 15.2 % for C4, 27.7 % for C6, 58.5 % for C8, and 57.3 % for C10. This discrepancy in efficiency can be attributed to higher C–F bond energy and increased hydrophilicity in short-chain PFAS [35,60]. In the case of PFOS, a common PFAS with a sulfonic group, the CeO_2 electrode achieved a desulfurization efficiency of 81.2 %, but a lower defluorination efficiency, slightly surpassing 60 % (Figure S20), suggesting that the initial reaction step may predominantly involve sulfonic acid group removal. It is crucial to note that many PFAS degradation studies typically use high concentrations ranging from 10 to 100 mg L^{-1}, which may overestimate the efficiency of the degradation process and make it challenging to accurately extrapolate these results to real-world scenarios, where PFAS concentrations are typically below 100 μg/L [17,61]. To more accurately replicate environmental conditions, our tests were conducted at initial PFAS concentrations of 1000 μg/L, 100 μg/L, and 10 μg/L. Remarkably, our process effectively removed PFAS across all tested concentrations, achieving removal efficiencies of 85 % at both 1000 μg/L and 100 μg/L, and 69 % at 10 μg/L. These results highlight the efficacy of the CeO_2 electrode in treating PFAS-contaminated waters, including at environmentally relevant concentrations.

To evaluate the durability and long-term cycling stability of the CeO_2 electrode in treating PFAS, we conducted tests to monitor the selective PFOA removal efficiency across nine consecutive cycles. Fig. 5d illustrates that the CeO_2 electrode consistently achieved approximately 80 % removal efficiency of PFOA over chloride ions. This stable performance across multiple cycles, facilitated by the successful electrode regeneration under reversed bias, demonstrates its sustained capacity for long-term operation. Additionally, the release of cerium remained less than 0.2 mg/L throughout the cycles (Figure S21), underscoring the robust performance and stability of the CeO_2 electrode with extended use.

To test this process in real environmental contexts, we used groundwater and surface water from the Shenzhen region, with water quality compositions detailed in Table S5. We prepared 1 mg/L PFOA aqueous solutions using these water samples and conducted PFOA removal experiments. The results, depicted in the Figure S22, showed PFOA removal efficiencies exceeding 60 % and defluorination efficiencies around 40 % for both types of water bodies. These rigorous validation experiments in complex water matrices confirm the application potential of this process for PFOA remediation.

The treatment of PFAS-contaminated water is often challenged by the high energy demands required to break the stable C–F bonds. We evaluated the energy consumption of our process using collected current data (Text S8, Figure S23), and compared it with established methods such as hydrothermal alkaline treatment (HALT), UV-based methods, plasma-based methods, and conventional electrochemical treatments. As illustrated in Fig. 6, while HALT, plasma, and UV-based methods can achieve PFAS mineralization efficiencies of over 80 %, these methods expose the entire volume of solutions to intense conditions, leading to substantial energy loss to the surrounding media and elevated consumption (200–800 kWh/m^3). These methods are effective for treating high-concentration PFAS wastewater but are less efficient for low-concentration solutions [62]. In contrast, conventional electrochemical methods use less energy (10–200 kWh/m^3) by focusing on more targeted approaches that minimize broad-spectrum energy losses. Nevertheless, these methods still require high oxidation potentials to break the robust C–F bonds, resulting in the oxidation of various species within the water, including other organic compounds, chloride ions, and water molecules. This contributes to their relatively high energy consumption despite a more targeted approach.

Our approach leverages the high affinity and unique interactions between Ce and fluorine-containing species, which facilitates adsorptive preconcentration and catalytic degradation of PFAS at low oxidation potentials. Additionally, the close proximity of the two electrodes generates a stronger electric field, reducing concentration polarization [4]. Under the same level defluorination efficiency of approximately 70 %, the treatment cost using CeO_2 electrodes is significantly reduced, reaching as low as 0.62 kWh/m^3 per order, far below that of other methods, including conventional electrochemical oxidation (Table S6). The key advantage of the CeO_2 electrode compared to traditional electrodes lies in its ability to achieve catalytic degradation at low potentials. This avoids the oxidation of chloride ions and NOM present in environmental media, eliminating the need for high concentrations of sodium sulfate electrolyte that are heavily relied upon in conventional electrochemical methods. These features make the CeO_2 electrode more suitable for real-world environmental treatment scenarios, ensuring improved scalability and robustness.

4. Conclusion

In this study, we demonstrated the efficient degradation of PFOA and other PFAS compounds at low potentials (1.37 V/SHE) using CeO_2 electrodes based on an affinity adsorption-activation strategy. This method achieved 94.0 % PFOA removal and 73.0 % defluorination at environmentally relevant concentrations, offering a significant improvement in PFAS removal efficiency. The adsorptive preconcentration and catalytic degradation of our CeO_2 electrodes allows for targeted adsorption and degradation of PFAS molecules, even at minimal concentrations, reducing the likelihood of contaminants escaping treatment and entering ecosystems. Importantly, successful trials in environmental matrix from groundwater and surface water in the Shenzhen region highlight the practicality and effectiveness of this technique, confirming its suitability for broader environmental remediation applications. The method was also validated across different PFAS types, concentrations, and water bodies, demonstrating its versatility.

With low energy consumption (0.62 kWh/m^3 per order), this approach is more cost-effective than many existing technologies, making it viable for large-scale environmental use. Both experimental and computational analyses confirmed the critical role of CeO_2 with oxygen vacancies in activating the C–F bond and lowering the energy barriers for the efficient defluorination of PFOA. These insights into the activation mechanisms provide a foundation for optimizing CeO_2-based electrodes and other materials for enhanced PFAS degradation and could extend to other persistent pollutants. Overall, the introduction of the low-voltage CeO_2 electrode method not only enhances PFAS treatment efficiency but also broadens the existing toolkit for combating PFAS pollution. While emphasizing treatment efficiency and cost, it is equally important to consider the realism of the environment. This involves not only accounting for background environmental factors but also addressing the concentration ranges and specific types of pollutants

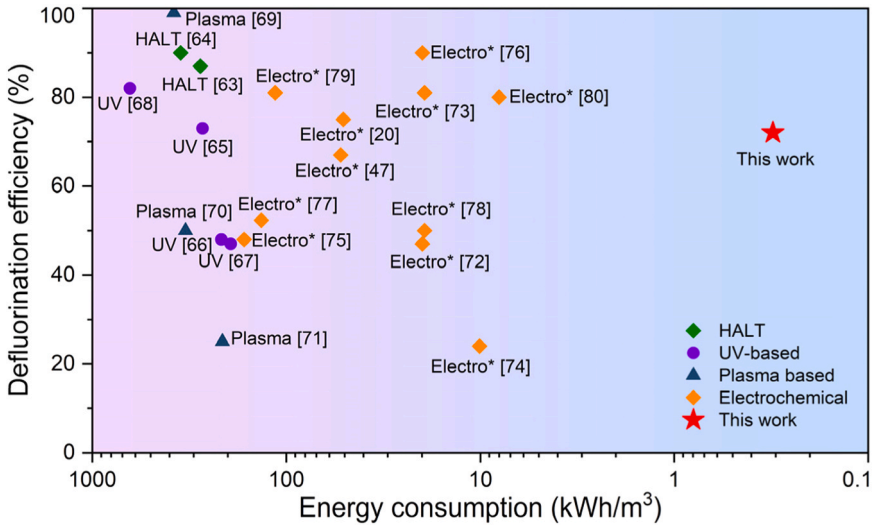


Fig. 6. Energy consumption comparison with other relevant technologies for degrading PFAS, including HALT [63,64], UV based [65–68], plasma based [69–71] and conventional electrochemical methods [20,47,72–80].

present in real-world scenarios. Such considerations are essential for developing practical and effective treatment technologies that can be applied in real environmental conditions. By addressing these aspects, research can bridge the gap between laboratory studies and field applications, paving the way for scalable, robust, and impactful solutions.

Environmental implication

The low-voltage CeO_2 electrode method improves the efficiency of PFAS degradation, particularly in scenarios with low pollutant concentrations. This approach, which integrates adsorptive concentration with catalytic degradation, enhances the removal process while addressing the economic and energy constraints associated with traditional methods. Demonstrated success in environmental matrix samples from Shenzhen validates its practical applicability. Further investigations into the role of oxygen vacancies in CeO_2 electrodes provide direction for future research aimed at refining materials and designs to boost the degradation efficiency of PFAS and other pollutants.

CRediT authorship contribution statement

Youmei Xu: Methodology, Investigation. **Chaoqun Zhu:** Methodology. **Mengxia Wang:** Methodology. **Aling Wan:** Methodology. **Bei Liu:** Writing – review & editing, Methodology. **Qi Han:** Methodology, Data curation. **Zhongying Wang:** Writing – review & editing, Supervision, Funding acquisition. **Xun Liu:** Methodology, Investigation, Data curation, Writing – original draft, Writting. **Yu Pan:** Investigation, Formal analysis. **Yufei Shu:** Methodology, Investigation, Data curation. **Meng Zhang:** Investigation, Formal analysis. **Guoshen Zeng:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2024.137043.

Data availability

Data will be made available on request.

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